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REVIEW ARTICLE

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Current challenges and future directions for institutional repositories: A systematic literature review. An Annual Review of Information Science and Technology (ARIST) paper

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Abstract

Institutional repositories (IRs) are essential in advancing Open Access and facilitating the dissemination of scholarly work. This systematic review examines the challenges faced by IRs in areas such as strategic alignment, content acquisition, funding, staffing, quality assurance, and technology. The review identifies recurring challenges. Strategic issues arise from the interaction between IRs, research organizations, and funders. Professionalization challenges encompass staffing, governance, and collaboration between libraries and IT departments. Funding continues to be a global issue, often tied to institutional resources and policy developments. Technological challenges include adapting to evolving tools while managing limited resources. Concerns related to content quality and copyright affect IR operations and researcher engagement. Usage challenges revolve around raising the awareness of researchers, understanding the role of IRs, and improving engagement through effective marketing and metrics. Key findings underscore the importance of strategic integration with institutional goals and funders' policies, professionalization and technological readiness, securing consistent funding, and addressing challenges in content acquisition, rights management, and researcher participation. This study contributes to the broader discourse on strengthening IRs as critical components of scholarly communication within the Open Science ecosystem.

1 | INTRODUCTION

Repositories are the organizational and technological information infrastructure for information management processes in many research performing organizations (RPOs). They aim to ensure the long-term documentation and accessibility of scientific publications and other materials generated during academic work.

In 1991, the Open Access repository arXiv was established at Los Alamos National Laboratory, providing open access to preprints, particularly in the field of high-energy physics in its early days (Butler, 1999; Ginsparg, 1994). Similar repositories were developed in other disciplines, such as CogPrints in cognitive science (Green, 2000) and NCSTRL in computer science (Davis, 1995). In 1993, CERN established one of

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the first institutional repositories (IRs) worldwide (Rodrigues Silvestre, 2010).

Repositories have been assigned a central role in the context of Open Access. As early as the Budapest Open Access Initiative of 2001 (Budapest Open Access Initiative, 2002), the term "open electronic archives" was used to underscore the critical role of repositories in facilitating the dissemination of freely accessible scholarly literature. Subsequent statements, including the Bethesda Statement on Open Access Publishing (2003) and the Berlin Declaration on Open Access to Knowledge in the Sciences and Humanities (2003), highlight the importance of the "online repository" as a key infrastructure for the implementation of green Open Access. Green Open Access enables free access ("gratis Open Access") to scholarly publications by allowing authors to deposit their work in repositories. This model helps to overcome financial barriers to accessing literature. However, technical and legal restrictions, such as limitations on text and data mining or reuse rights, often persist, which may inhibit the reuse of publications (Suber, 2012).

A widely accepted definition of an IR can be found in Lynch (2003). The author defines an IR as "a set of services that a university offers to members of its community for the management and dissemination of digital materials created by the institution and its members. It is most essentially an organizational commitment to the stewardship of these digital materials, including long-term preservation where appropriate, as well as organization and access or distribution." This definition highlights that operating repositories is an organizational and technical service provided by information infrastructure institutions for a defined community of researchers. In Europe, most repositories are funded by the institutions that operate them (Shearer et al., 2023).

With the advent of software solutions such as DSpace (Smith et al., 2003), E-Prints (Carr, 2010), and OPUS (Scholze, 2004), academic libraries at RPOs began to promote electronic publishing. IRs gained importance at universities, particularly towards the end of the 20th century, for publishing theses and dissertations (Fox et al., 2002; Yiotis, 2008). IRs also advanced the concept of "distributed, institution-based self-archiving," as described by (Harnad, 2001). Harnad envisioned that researchers, supported by libraries, would store their publications in repositories, thereby fostering the green route to Open Access (Bosc & Harnad, 2005). Thanks to the Open Archives Initiative Protocol for Metadata Harvesting (OAI-PMH), published in 2001 (Lagoze & Van de Sompel, 2001), distributed publications and metadata can be aggregated and made discoverable via search engines, such as the Bielefeld Academic Search Engine (BASE) (Lossau, 2004). Various standardization initiatives, such as the "DRIVER Guidelines" (Vanderfeesten et al., 2008) and the "OpenAIRE Guidelines" (Schirrwagen & Baglioni, 2018), focus on improving standards and interfaces.

Several studies document the growing number of publications available in IRs (Gargouri et al., 2012; Piwowar et al., 2018). According to OpenDOAR, as of January 2025, there are 5983 Open Access Repositories worldwide.[1] When making journal articles accessible, particularly those already published in closed-access journals, the relevant copyright provisions, documented in a Copyright Transfer Agreement (CTA), must be considered. Most journals allow free access through repositories, sometimes requiring an embargo period or other conditions (Laakso, 2014). A significant challenge is the reluctance of many authors to engage in self-archiving (Creaser et al., 2010). With growing demands for research reporting and the associated professionalization of publication management, IRs and Current Research Information Systems (CRIS) have increasingly been considered complementary to each other (Biesenbender et al., 2019; Jeffery & Asserson, 2009). Since 2013, a second wave of preprint server establishment has been observed (Callaway, 2013). A prominent actor in this wave is the repository bioRxiv in the biosciences (Abdill & Blekhman, 2019). This trend became especially evident during the COVID-19 pandemic, when preprint servers saw significant usage (Fraser et al., 2021).

Distinctions between institutional or subject-specific repositories, publication databases, or journal platforms are increasingly fading due to evolving technologies and the growing ubiquity of Open Access. This is exemplified by the vision of the Confederation of Open Access Repositories (COAR) of "Next Generation Repositories" (Rodrigues et al., 2017). This vision aims to transform repositories into central, open, and interconnected infrastructures for Open Science through interoperability, innovative functionalities, and global integration. With the growing political emphasis on Open Science (UNESCO, 2021), there is an increasing trend to make not only texts but also research data and scientific software accessible through repositories. While some RPOs operate specific IRs for certain publication types or recommend the use of disciplinary and specialized repositories (e.g., domain repositories for research data), others operate generic repositories, independent of discipline and publication type, where all products of scholarly work can be made accessible in the spirit of Open Science.

The role of repositories in scholarly communication is significantly influenced by policies from Research Funding Organizations (RFOs). In the United States, repositories play a crucial role in implementing the so-called Nelson Memo (White House Office of Science and Technology Policy (OSTP), 2022). In Europe, the European Commission mandates the accessibility of publications

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produced under the ninth framework program Horizon Europe through repositories (European Commission, 2024). However, the degree of repository integration varies between European countries (European Commission, 2020). In Japan, the relevant ministry announced in 2024 plans to invest 10 billion Yen in expanding IRs (Singh Chawla, 2024).

While the Open Access debate in some countries has focused heavily on so-called transformative agreements (Rothfritz et al., 2024), repositories continue to play an important role in this landscape. Services such as the Publication Router in the United Kingdom (UK) (Burland & Grout, 2017) and DeepGreen in Germany (Stompor et al., 2025), for example, automatically deliver publications produced under transformative or gold Open Access agreements to IRs. This workflow ensures that articles from hybrid journals as well as fully gold Open Access journals are also archived in IRs.

In light of the developments outlined here, IRs face new challenges constantly. IRs interact with a wide range of developments. External factors, including RFO policy developments and technological advances, impact the operations and strategies of IRs within RPOs, presenting significant challenges. However, internal developments also play a critical role. In a study focusing on European repositories, three main challenges were emphasized in relation to their operation: (1) Software upgrades were highlighted as the main challenge, showcasing the continuous need to update and maintain technical infrastructure to keep repositories functional and secure. (2) Skilled staff was the second most cited issue, stressing the importance of having well-trained and knowledgeable staff for efficient management of repository operations. (3) Underfunding was cited as the third challenge; underfunding underscores the financial difficulties in sustaining and improving repository services. These challenges illustrate the complex landscape of managing repositories (Shearer et al., 2023).

On an international level, COAR and Scholarly Publishing and Academic Resources Coalition (SPARC) have developed a strategic vision for repositories through community consultation (Baich, 2022). The understanding of the two associations is that repositories play a crucial role in national research infrastructures by ensuring that research outputs are not only equitable and inclusive, but also discoverable, visible, and accessible for future use. They view repositories as institutional tools that support the mission of disseminating research and provide opportunities for innovative publishing methods (SPARC, n.d.).

Against this backdrop, this study initiates an inquiry into the current state of research on the challenges associated with managing IRs. A focus on IRs as a type of repository has been chosen because this type is the most widespread and has been best described in the field of Library and Information Science (LIS) to date. This systematic literature review provides a comprehensive description of the challenges faced by IRs, as discussed in English-speaking LIS research literature. Considering the dynamic landscape in which IR systems operate, this study explores the challenges they face and how they address these challenges.

2 | METHODOLOGY

With this literature review we attempt to identify, extract, group, and summarize relevant challenges for the global repository community in a comprehensive way, methodically and systematically. This study was carried out according to the statement Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) and the PRISMA protocol (Page et al., 2021). The main subject area of the study is LIS. The review was conducted by four LIS scientists who are members of the Information Management research and teaching group at the Berlin School for Information Science at Humboldt-Universität zu Berlin, Germany, as part of the Pro OAR DE project (Pampel & Rothfritz, 2023).

2.1 | Research question

Institutional repositories have to meet the recent expectations of researchers, research institutions, and libraries in order to become "globally networked infrastructure for scholarly communication" (Shearer et al., 2023). However, there are still a variety of issues that repositories face. We assume that these issues are most likely spread across a continuum of different dimensions and can range from technical to organizational requirements that have not yet been met. They may also change according to locality and development status, as well as governmental or organizational support for a repository. With this in mind, we approached this review with the following research question:

> **RQ1.** *What are key challenges for the operation of institutional repositories according to literature from Library and Information Science?*

2.2 | Search approach

For this systematic review, we used Scopus, the scientific abstract and citation database by Elsevier, and the subject-specific database "Library and Information

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Science Abstracts" (LISA) by ProQuest, which, as a tool for abstracting and indexing scholarly literature on LIS, covers over 440 periodicals from more than 68 countries and in more than 20 different languages.[2]

We used a broad search approach by searching for the keywords "Repository" and "Repositories." The approach yielded 25,350 records. We then refined the results by applying filters offered by LISA. We applied the following filters:

- Limit to: Peer Reviewed
- Publication Date: January 1st, 2014–December 31st, 2023
- Subject: Institutional Repositories
- Language: English, German

This resulted in a sum of 934 records in total.

Using Scopus, we searched for the same keywords (i.e., "repository" and "repositories"), which resulted in 107,067 records. We then applied the following filters from the Scopus interface:

- Year: 2014–2023
- Language: Limited to English, Limited to German
- Document type: Article
- Keyword: Limited to Institutional Repository

The search string for Scopus thus looked like this:

```
TITLE-ABS-KEY ( repository ) AND PUBYEAR > 2014 AND
PUBYEAR 2024 AND
( LIMIT-TO ( EXACTKEYWORD, "Institutional
Repositories") )
AND ( LIMIT-TO ( LANGUAGE, "english" ) )
AND ( LIMIT-TO ( LANGUAGE, "german" ) )
AND ( LIMIT-TO ( DOCTYPE, "ar" ) )
```

This resulted in a sum of 334 records in total.

2.3 | Inclusion and exclusion criteria

To ensure the relevance and conceptual focus of this systematic review, we applied a set of exclusion criteria designed to filter out publications that did not align with the objective of the study: to synthesize evidence on challenges related to the development, management, and strategic integration of IRs. Records were excluded if they primarily addressed topics outside this scope, such as purely technical descriptions of repository platforms or narrowly focused case studies that did not extrapolate to broader organizational or infrastructural issues. Similarly, publications whose main emphasis was on patterns of IR use by researchers or students were excluded, as these studies, while valuable, did not systematically examine institutional, policy, or operational challenges.

TABLE 1 Exclusion criteria.

Exclusion criteria	No. of excluded records
Excluded because other topics than issues, trends, policies, services were addressed	250
Excluded because of wrong document type (e.g., book review, editorial, opinion paper, statement, specific (single) use case)	134
Excluded because of a singular focus on IR usage by researchers or institutional members	53
Excluded because of a singular focus on repository technology	42
Excluded because of a focus on other types than "institutional repository" (e.g., "archive," "disciplinary repository," "learning repository")	24
Excluded because full text was not available or no English or German version was available	22
Excluded because the record addressed the wrong setting (not a research institution)	4

Additionally, to maintain methodological rigor and comparability, we excluded document types that typically do not present empirical or comprehensive conceptual analyses, such as editorials, opinion pieces, book reviews, and brief statements, as well as records where the full text was unavailable or not in English or German, which limited the ability to assess study quality and relevance. We also excluded studies focusing exclusively on other types of repositories, such as disciplinary or generalist repositories, to avoid conflating challenges unique to institutional contexts with those emerging from domain-specific or pedagogical settings. These criteria were applied iteratively during the screening process, as detailed in Table 1, and helped establish a coherent and analytically comparable corpus for the review.

The filtered records and metadata ($N = 934$) were extracted on January 23, 2024, using the LISA database and exported as RIS files (including reference to the literature, abstract, indexing information) to Zotero. We first screened the abstracts of all 934 included documents. This screening was carried out as a group and led to the inclusion of 515 documents for further in-depth evaluation. Four-hundred and nineteen documents were excluded from retrieval because they did not meet the inclusion criteria according to their abstracts. All remaining documents were in English, none in German. We

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exported the remaining sample ($n = 515$) to a separate Zotero folder, retrieved the PDFs of the remaining 515 records, and started screening the sample. All three authors screened through the full text and discussed their decisions as a team. Four-hundred and eighty-one records were excluded from the sample in the final step of screening. The exclusion criteria are described in Table 1. We included studies and research articles in English or German that focused on IRs and identified or addressed challenges and issues from the point of view of repository managers or other information professionals. The final sample included in the review consisted of 34 records in total.

During the revision process, we expanded our search to include Scopus. Records were identified from Scopus on March 17, 2025. After applying search filters, 334 papers were exported from Scopus. To identify duplicates between databases, we first cross-checked Digital

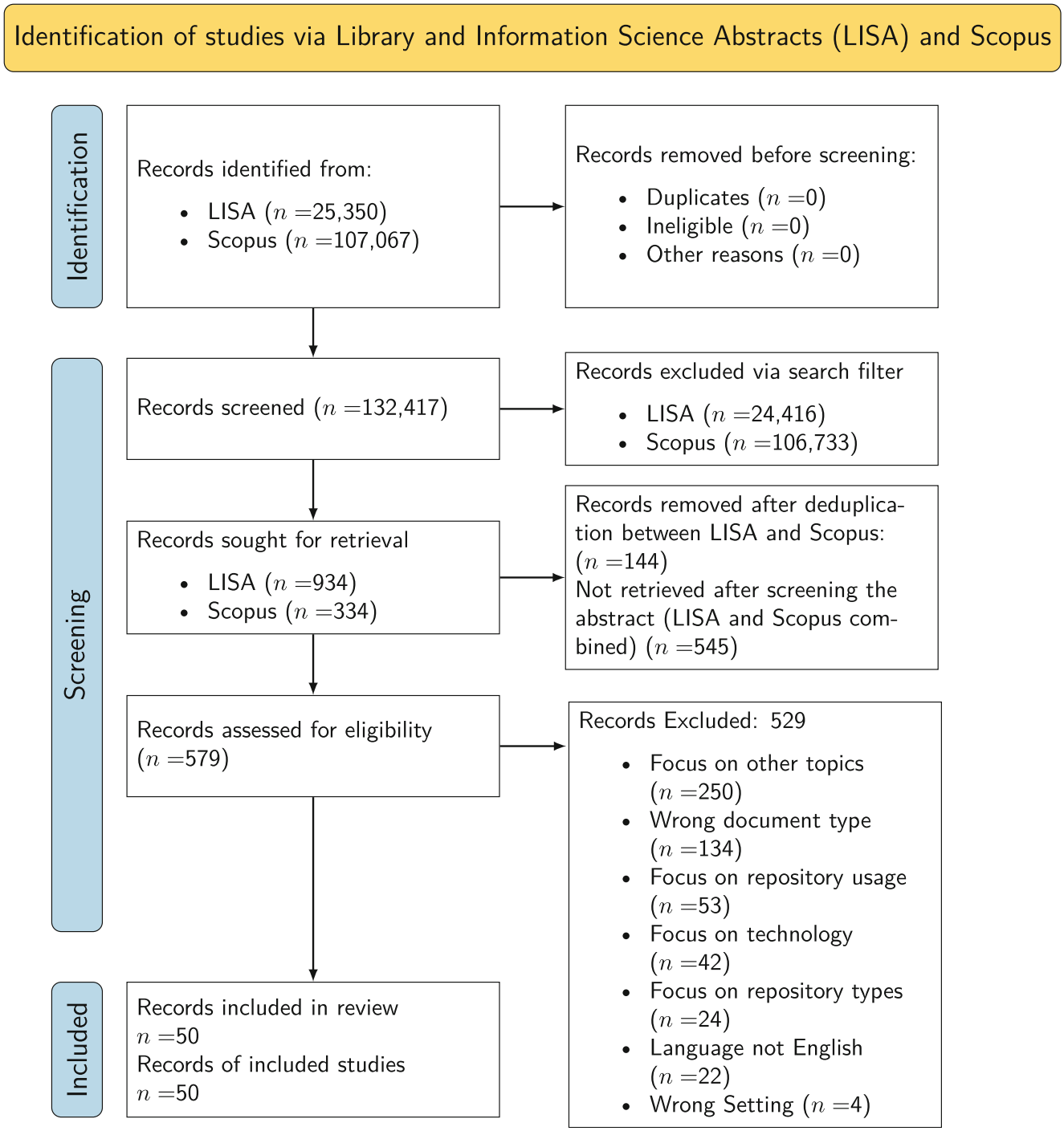


FIGURE 1 PRISMA 2020 statement and workflow diagram. Diagram adapted from Page et al. (2021).

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Object Identifiers (DOIs), and for papers without DOIs, we checked titles and authors. This process identified 190 new records that were not present in the LISA database. After screening abstracts, 126 records were excluded, leaving 64 records for full-text eligibility assessment. Of these, 48 were excluded according to the criteria listed in Table 1, resulting in 18 records from Scopus included in the review. The final sample included in the review consisted of 50 records in total (34 from LISA and 16 from Scopus).

2.4 | PRISMA compliance

We conducted the systematic review in accordance with the PRISMA protocol. The full set of records, including filter, inclusion criteria, and exclusion criteria, is documented in the PRISMA 2020 flow diagram (see Figure 1).

2.5 | Data extraction

The full texts for all records in the final sample ($n = 50$) were analyzed applying computer-assisted qualitative methods in the form of inductive in-depth coding using the MAXQDA software (version 24.6.0). The coding was performed by three authors independently and discussed in the group.

We also checked all articles in the final sample for the following information:

- Methodological approach: Did the studies use quantitative, qualitative, or mixed methods?
- Methods: What type of methods were used to collect the data?
- Population or sample and response rate: Where were the data collected? How large was the sample size (N) and how high was the response rate (n)?

This methodological check was important to exclude purely descriptive publications, comments, or other opinion pieces, or publications in which the method was not clearly described from the sample.

Lastly, we enriched the bibliographic data via the OpenAlex API[3] by OurResearch (Priem et al., 2022) to determine additional publication information, as well as to cross-check origin information included in the metadata. We used OpenRefine, version 3.7.2, to retrieve all metadata available for each item from the publicly available Application Programming Interface (API). We then extracted information on Open Access status and type, repository availability, and affiliation country for the first authors. We were able to do this for 41 articles in the sample. The remaining nine articles did not include a DOI; automatic metadata retrieval was therefore not possible.

3 | RESULTS

3.1 | Overview

The final sample included 50 articles published between 2014 and 2024. The number of articles per year varied between one in 2014 and 2022, respectively, and ten in 2019. There is no distinct upward or downward trend; however, the sample shows a maximum number of publications in 2019 (10) (see Figure 2).

The articles included in the sample show the following continental distribution based on the first authors' affiliations (see Figure 3).

Most of the included articles (50%, $n = 25$) originated on the African continent. Twenty-two ($n = 11$) percent of articles originated in North America, 18% ($n = 9$) in Asia, 8% from Europe ($n = 4$), and 2% ($n = 1$) in South America. No article originated from Oceania. The local distribution at the country level showed the majority of articles with the first authors affiliated with Nigeria (30%, $n = 15$) and the United States (US) (24%, $n = 12$) (see Table 2).

Of the 50 articles included in the sample, 41 included a DOI in their metadata. The DOI allowed us to check for publication status in the bibliographic metadata. Of 41 articles on IRs, 27 (66%) were published as closed

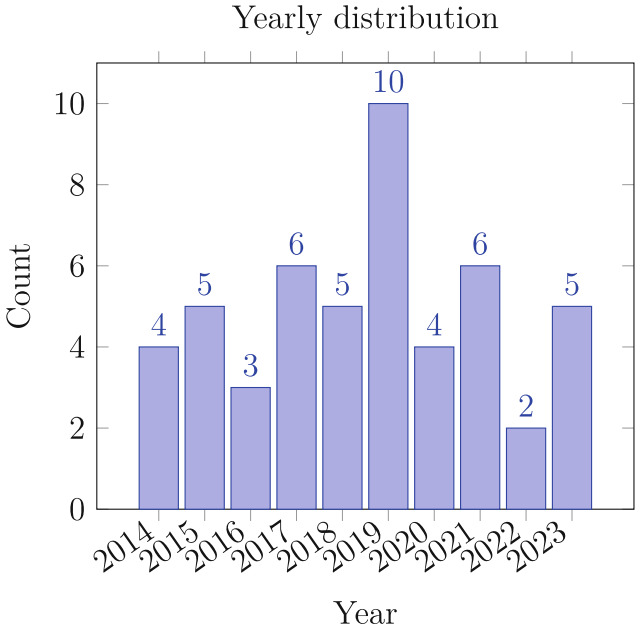


FIGURE 2 Number of articles per year, $n = 50$.

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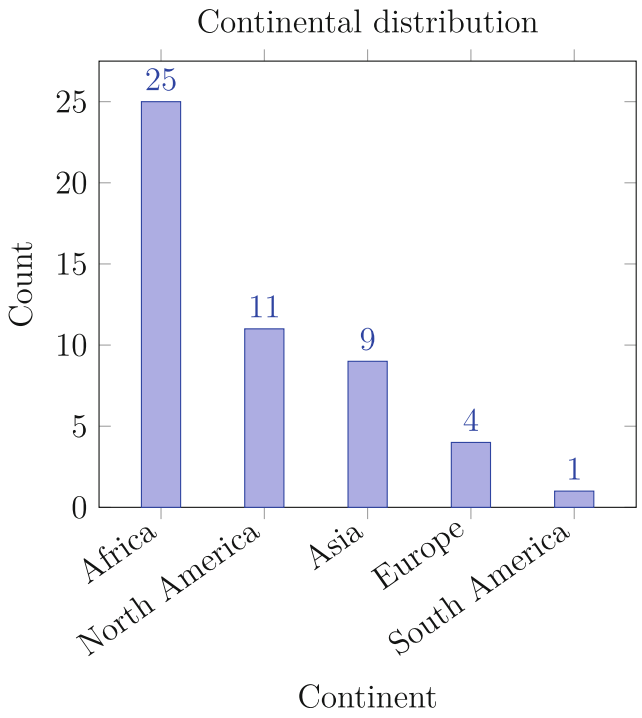


FIGURE 3 Number of articles with first author affiliation per continent, $n = 50$.

access and 14 (34%) were published as Open Access articles, as illustrated in Figure 4.

Out of the 14 Open Access articles, five were published via Gold Open Access in Diamond Open Access journals, seven via Green Open Access (two of these were published in Gold Open Access journals and one was published in a Diamond journal), four via Gold Open Access in a Gold Open Access journal (one of these was a Bronze Open Access article), and one article was published as a Hybrid Open Access article. Only seven of the 14 Open Access articles and seven out of 41 articles in total could be located in a repository (see Figure 5).

The methodological approaches in our samples varied, but most of the studies employed quantitative methods (22 articles, 44%) followed by mixed methods approaches (15 articles, 30%) and finally qualitative methods (12 articles, 24%). The approach for one article was not classifiable. Figure 6 shows the distribution of methodological approaches among the sample.

In terms of quantitative methods, surveys, infometrics, and descriptive statistics were most prevalent. The surveys ranged from larger scale, for example, for all researchers at five universities in Ghana (Kodua-Ntim & Fombad, 2020) to small scale, for example, single representatives of 23 IR in South India (Shajitha & Majeed, 2021). The response rates were high in most articles that described results from surveys. They reached 20% in all studies, except one, and up to 97.22% in some. Descriptive statistics and infometrics were used in

TABLE 2 Country of affiliation for first authors according to bibliographic metadata.

Country	No. of papers	References
Nigeria	15	Anenene et al. (2017), Anyaoku et al. (2019), Baro and Nwabueze-Echedom (2023), Baro and Kari (2016), Baro et al. (2014), Doris and Ekereuche (2023), Ejikeme and Ifeanyi (2019), Fasae et al. (2017), Oberhiri-Orumah and Baro (2023), Ifijeh et al. (2018), Okoroma (2018), Utulu and Ngwenyama (2021), Uba and Okonkwo (2021), Usman and Kaur (2023), and Ukwoma et al. (2019)
US	12	Ammarukleart and Kim (2017), Baughman et al. (2018), Carlson (2015), Dawson and Yang (2016), Dubinsky (2014), Hwang et al. (2020), Joo et al. (2019), Kipnis et al. (2019), Quinn (2023), Scherer and Valen (2019), Sterman (2014), and Zhong and Jiang (2016)
Ghana	3	Adjei et al. (2019), Kodua-Ntim and Fombad (2020), and Martin-Yeboah et al. (2018)
UK	3	Marsh (2015), Pinfield et al. (2014), and Ten Holter (2020)
India	2	Mahanta (2021) and Shajitha and Majeed (2021)
Indonesia	2	Farida et al. (2015) and Liauw and Genoni (2017)
Malaysia	2	Adam and Kiran (2021) and Asadi et al. (2019)
Pakistan	2	Rafiq (2022) and Ullah and Mirza (2020)
Uganda	2	Kakai (2021) and Kakai et al. (2018)
Botswana	1	Oladokun (2015)
Cuba	1	Martí-Lahera and Puerta-Díaz (2019)
Eswatini	1	Dlamini and Snyman (2017)
Greece	1	Kyprianos and Lygnou (2022)
Kenya	1	Chilimo (2015)
Sudan	1	Abdelrahman (2017)
Zimbabwe	1	Tapfuma and Hoskins (2019)

several articles to describe repository infrastructures and their content (Farida et al., 2015; Pinfield et al., 2014) and were also often mixed with another method, either a survey (Dlamini & Snyman, 2017; Oberhiri-Orumah & Baro, 2023; Shajitha & Majeed, 2021) or an interview study for a mixed methods approach (Chilimo, 2015; Zhong & Jiang, 2016). Other mixed methods included mostly a combination of interviews and surveys

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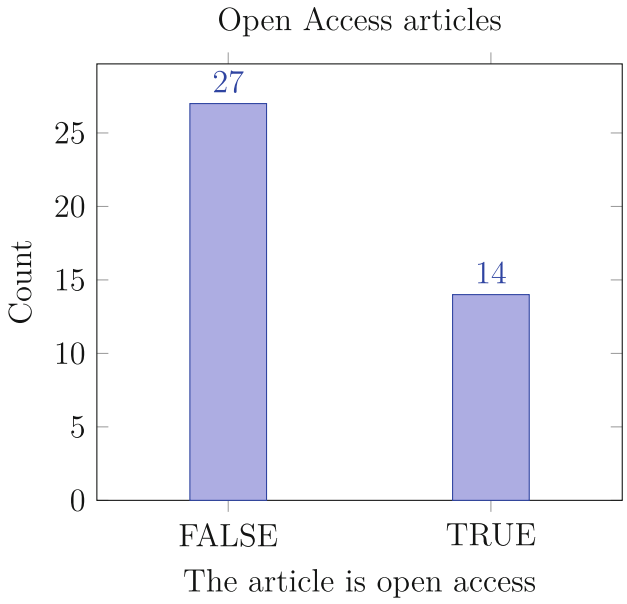


FIGURE 4 Number of articles published as Open Access, $n = 41$.

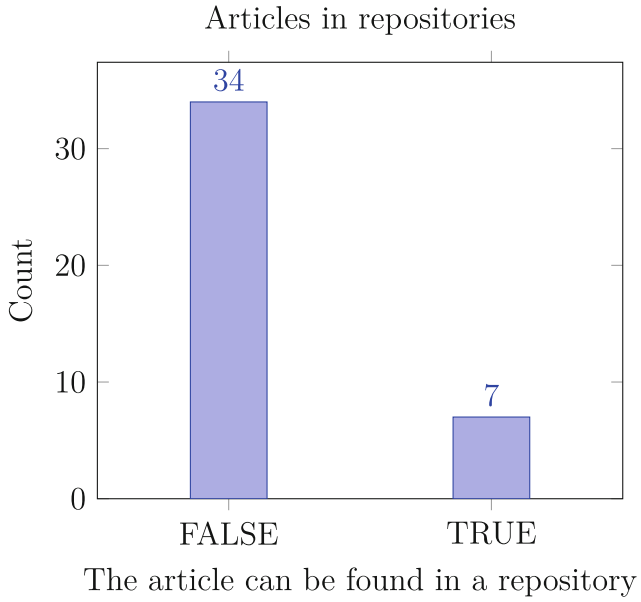


FIGURE 5 Number of articles published in repositories, $n = 41$.

(Kakai, 2021; Kakai et al., 2018; Kodua-Ntim & Fombad, 2020; Uba & Okonkwo, 2021), interviews and focus groups (Rafiq, 2022), or observations (Utulu & Ngwenyama, 2021). Qualitative methods varied between document and content analysis (Dawson & Yang, 2016; Liauw & Genoni, 2017; Martí-Lahera & Puerta-Díaz, 2019), interviews (Adam & Kiran, 2021; Adjei et al., 2019; Marsh, 2015; Quinn, 2023; Ten Holter, 2020), or case studies (Martí-Lahera & Puerta-Díaz, 2019; Scherer & Valen, 2019).

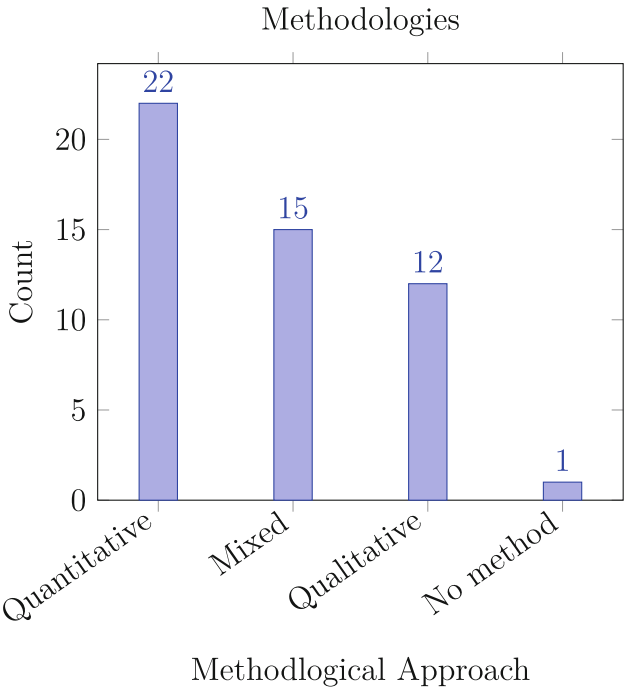


FIGURE 6 Types of methodological approaches for all articles in the sample, $n = 50$.

Inductive coding of all articles yielded 15 codes in total. All codes describe a domain in which challenges for IRs were described. During this process, the codes shown in Table 3 emerged.

In all articles included in the sample, at least one of the codes was present. Table 4 provides an overview of the distribution of all codes in the sample.

For subsequent analysis, we clustered the codes into six overarching thematic topics. The following sections present an in-depth discussion of the results for each topic.

3.2 | Strategic anchoring

Strategic anchoring in the context of IRs refers to the alignment of repositories with overarching institutional priorities and standards. This includes the establishment and enforcement of clear policies and mandates, the promotion of institutional commitment at the leadership and operational levels, and compliance with internal and external requirements and standards. Challenges in this domain were found in segments coded with "Policies, mandates" "Institutional commitment," or "Compliance."

Of the 50 articles in the sample, 39 articles (78%) contained challenges with respect to strategic anchoring. Articles that did not mention strategic anchoring were articles that were more concerned with technical

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TABLE 3 Inductive codes.

Code	Description
Compliance	Coded for instances when challenges with compliance with policies or mandates were mentioned
Data sharing	Coded when when challenges with data sharing in repositories were mentioned
Financing, funding	Coded when challenges with financing or funding repositories were mentioned
Impact, statistics	Coded when challenges with measuring impact of repository content via statistics were mentioned
Institutional commitment	Coded when challenges regarding commitment to the repository by institutional leadership was mentioned
Marketing, communication	Coded when challenges with marketing or communicating a repository's values were mentioned
Metadata	Coded when challenges with metadata collection or management were mentioned
Policies, mandates	Coded when challenges due to external policies and mandates, or lack thereof, where mentioned
Preservation	Coded when when challenges with preserving digital records in a repository were mentioned
Quality, reputation	Coded when challenges with content quality or reputation mechanisms through repository content were mentioned
Repository governance	Coded when challenges with repository governance and workflows were mentioned
Rights	Coded when challenges with rights management in repositories were mentioned
Staff	Coded when challenges regarding repository staffing were mentioned
Technology, Infrastructure	Coded when when challenges regarding technical infrastructures were mentioned
Usage, awareness	Coded when challenges with researchers' awareness and usage of a repository were mentioned

solutions (Baughman et al., 2018; Carlson, 2015; Mahanta, 2021; Scherer & Valen, 2019), legal challenges (Tapfuma & Hoskins, 2019), awareness (Marsh, 2015), marketing and financing (Utulu & Ngwenyama, 2021), staff and financing (Baro & Kari, 2016), mentioned several of these challenges (Adam & Kiran, 2021; Anyaoku et al., 2019), or offered a high-level overview of the topic (Ammarukleart & Kim, 2017).

All of the 39 articles mentioning challenges with regard to strategic anchoring were coded with "policies, mandates," except for five articles (Abdelrahman, 2017; Anenene et al., 2017; Hwang et al., 2020; Kakai, 2021; Oladokun, 2015). The development and implementation of policies governing IRs show significant variability between institutions and regions. This is apparent in many articles in our sample that cover countries throughout the African continent, such as Nigeria (Baro et al., 2014; Doris & Ekereuche, 2023; Ejikeme & Ifeanyi, 2019; Ifijeh et al., 2018; Okoroma, 2018; Uba & Okonkwo, 2021; Ukwoma et al., 2019; Usman & Kaur, 2023), Ghana (Adjei et al., 2019; Kodua-Ntim & Fombad, 2020; Martí-Lahera & Puerta-Díaz, 2019), or Kenya (Chilimo, 2015; Kakai, 2021; Kakai et al., 2018). The absence of clear and comprehensive institutional policies has been a persistent challenge. The results show that while some universities have successfully drafted IR policies, these often lack mandatory provisions, making the deposit processes voluntary and less effective (Baro et al., 2014; Baro & Nwabueze-Echedom, 2023; Doris & Ekereuche, 2023; Kakai et al., 2018; Oberhiri-Orumah & Baro, 2023; Uba & Okonkwo, 2021). For example, in a survey conducted with Nigerian university libraries, 92% of respondents agreed or strongly agreed that the lack of IR policies challenged the development of IRs in Nigeria (Oberhiri-Orumah & Baro, 2023).

But challenges around policies and mandates also extend to countries on other continents represented in the sample. Mandates vary between centralized and decentralized approaches. For example, in the UK, the development of distributed institution-based repositories has been largely mandated by infrastructure programs like Jisc, while France has opted for centralized repositories, such as HAL (Pinfield et al., 2014). Although mandatory deposit policies are recognized as critical to increasing repository usage (Ifijeh et al., 2018), their impact depends on the presence of compliance monitoring and associated incentives or sanctions (Chilimo, 2015). Pinfield et al. (2014) emphasize that policy mandates must be clearly worded and actively enforced to achieve widespread adoption. However, many RPOs, particularly in developing regions, struggle to implement such measures due to limited resources and bureaucratic obstacles (Abdelrahman, 2017; Baro et al., 2014; Baro & Nwabueze-Echedom, 2023; Kakai et al., 2018; Uba & Okonkwo, 2021; Ullah & Mirza, 2020).

Compliance with Open Access policies, particularly those of RFOs, presents another set of challenges in this area. These challenges present themselves in two ways, referring to either challenges with compliance by researchers or challenges with monitoring compliance by IR managers. If there are Open Access policies, ensuring compliance by researchers can often be challenging. This point is illustrated by Ten Holter (2020), whose study

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TABLE 4 Code distribution for each article.

	Compliance	Data sharing	Financing, funding	Impact, statistics	Institutional commitment	Marketing, communication	Metadata	Policies, mandates	Preservation	Quality, reputation	Repository governance	Rights	Staff	Technology, infrastructure	Usage, awareness
Abdelrahman (2017)	0	0	1	0	1	0	0	0	0	0	0	0	1	0	1
Adam and Kiran (2021)	0	0	1	0	0	0	0	0	0	0	0	1	1	1	0
Adjei et al. (2019)	0	0	1	0	0	0	0	1	0	0	0	0	1	1	0
Ammarukleart and Kim (2017)	0	1	0	0	0	0	0	0	0	0	0	0	0	1	0
Anenene et al. (2017)	0	0	0	0	1	0	0	0	0	0	0	1	0	0	0
Anyaoku et al. (2019)	0	0	1	0	0	0	0	0	0	0	0	1	1	1	1
Asadi et al. (2019)	0	0	1	0	1	0	0	1	0	0	0	1	1	1	1
Baro et al. (2014)	0	0	1	0	0	0	0	1	0	0	0	1	1	1	0
Baro and Kari (2016)	0	0	1	0	0	0	0	0	0	0	0	0	1		
Baro and Nwabueze-Echedom (2023)	0	0	1	0	0	0	0	1	0	0	0	1	1	1	1
Baughman et al. (2018)	0	0	0	1	0	0	0	0	0	0	0	0	0	1	0
Carlson (2015)	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0
Chilimo (2015)	1	0	0	1	0	0	0	1	0	0	0	0	0	1	0
Dawson and Yang (2016)	0	0	0	0	0	0	0	1	0	0	1	1	0	0	0
Dlamini and Snyman (2017)	1	0	1	0	1	0	0	1	0	0	1	0	0	1	1
Doris and Ekereuche (2023)	0	0	1	0	1	0	0	1	0	0	1	1	1	1	1
Dubinsky (2014)	1	1	0	1	1	0	0	1	0	1	0	1	0	0	1
Ejikeme and Ifeanyi (2019)	1	0	1	1	1	0	0	1	0	0	0	0	1	1	0
Farida et al. (2015)	1	0	0	0	1	0	0	1	0	1	0	1	0	0	1
Fasae et al. (2017)	0	0	1	0	1	1	0	1	0	0	1	1	1	1	1

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TABLE 4 (Continued)

	Compliance	Data sharing	Financing, funding	Impact, statistics	Institutional commitment	Marketing, communication	Metadata	Policies, mandates	Preservation	Quality, reputation	Repository governance	Rights	Staff	Technology, infrastructure	Usage, awareness
Oberhiri-Orumah and Baro (2023)	0	0	1	0	1	0	0	1	0	0	0	1	1	1	1
Hwang et al. (2020)	0	1	0	0	1	1	0	0	0	0	0	1	1	0	1
Ifijeh et al. (2018)	0	0	1	0	1	0	0	1	0	1	0	1	1	1	1
Joo et al. (2019)	0	1	1	0	1	1	1	1	1	1	1	1	1	1	1
Kakai et al. (2018)	1	0	0	0	1	1	0	1	0	0	0	1	1	1	1
Kakai (2021)	1	0	0	1	1	0	0	0	0	0	0	1	1	0	1
Kipnis et al. (2019)	0	0	1	1	1	0	1	1	0	0	0	1	1	1	1
Kodua-Ntim and Fombad (2020)	0	0	0	0	1	1	0	1	0	1	0	1	1	1	1
Kyprianos and Lygnou (2022)	0	0	0	0	0	0	0	1	0	0	0	1	1	0	0
Liauw and Genoni (2017)	0	1	0	0	0	1	1	1	1	1	1	1	0	0	0
Mahanta (2021)	0	0	1	0	0	0	0	0	1	0	0	0	1	1	0
Marsh (2015)	0	0	0	1	0	0	1	0	0	0	0	1	0	1	1
Martí-Lahera and Puerta-Díaz (2019)	0	0	0	1	1	0	1	1	0	0	0	0	0	0	0
Martin-Yeboah et al. (2018)	0	0	1	1	1	1	0	1	0	0	0	1	1	1	1
Okoroma (2018)	0	0	1	0	1	0	0	1	1	1	1	1	1	1	1
Oladokun (2015)	1	0	1	1	0	1	0	0	0	0	0	1	1	1	1
Pinfield et al. (2014)	1	0	0	1	1	0	0	1	1	1	1	0	0	1	1
Quinn (2023)	0	1	0	1	1	1	0	1	1	1	1	1	1	1	1
Rafiq (2022)	0	0	1	1	1	1	0	1	0	1	1	1	1	1	1
Utulu and Ngwenyama (2021)	0	0	1	0	0	1	0	0	0	0	1	0	1	1	0
Utulu and Ngwenyama (2021)	0	0	0	0	0	0	0	0	0	0	0	1	1		
Shajitha and Majeed (2021)	0	0	0	0	0	0	0	1	0	0	0	1	1	0	1
Sterman (2014)	1	0	1	1	1	1	0	1	1	0	1	1	1	1	1

(Continues)

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TABLE 4 (Continued)

	Compliance	Data sharing	Financing, funding	Impact, statistics	Institutional commitment	Marketing, communication	Metadata	Policies, mandates	Preservation	Quality, reputation	Repository governance	Rights	Staff	Technology, infrastructure	Usage, awareness
Tapfuma and Hoskins (2019)	0	0	0	0	0	0	1	0	0	0	1	1	1	0	0
Ten Holter (2020)	1	1	1	1	0	1	1	1	0	1	1	1	1	1	1
Uba and Okonkwo (2021)	0	0	1	0	0	0	0	1	0	1	0	1	0	1	0
Ukwoma et al. (2019)	0	0	1	0	1	0	0	1	0	0	0	0	1	1	1
Ullah and Mirza (2020)	0	0	1	0	1	0	0	1	0	0	0	1	1	1	0
Usman and Kaur (2023)	0	0	1	0	1	0	0	1	0	0	1	0	1	0	1
Zhong and Jiang (2016)	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1

Note: Binary view, in which 1 means code is present in article, 0 means code is not present.

describes how researchers deposit papers into the IR "begrudgingly" (p. 14) because it is mandated through the Research Excellence Framework (REF) in the UK. This study also points to the fact that compliance with REF is also a challenge for Open Access monitoring by IR managers. The interview data spotlight how, in the opinion of this population, mandates such as REF incentivize researchers to deposit their work into the IR but at the same time devalue its overarching function because neither the institutional leadership nor researchers see value in the IR outside of REF compliance. In some disciplines, it can also be very important to ensure that institutional reporting infrastructures and IR initiatives are not redundant but complement each other (Kipnis et al., 2019).

Institutional commitment is a critical factor for the strategic anchoring of IRs. Twenty-eight articles in the sample (56%) mentioned challenges in institutional commitment. The lack of institutional commitment can be understood in two ways: institutional management and leadership support are lacking, or there is a stark lack of awareness of why IRs should and do exist (Asadi et al., 2019; Dlamini & Snyman, 2017; Ejikeme & Ifeanyi, 2019; Fasae et al., 2017; Oberhiri-Orumah & Baro, 2023; Rafiq, 2022; Ullah & Mirza, 2020). It should be noted that this lack of awareness does not extend to user awareness in these articles, but to awareness by stakeholders responsible for decision-making at the RPO leadership level. For example, results from a survey among Pakistani universities and a subsequent focus group discussion with librarians described a lack of conceptual understanding and commitment by institutional administration as one of the major challenges for librarians in the development of IRs (Rafiq, 2022).

In summary, the interaction between IRs and the policies of RPOs and RFOs can be noted as a challenge. In the literature, the interaction between IRs and policies is discussed at various levels. Although clear policies and mandates can drive repository development, their absence or weak enforcement often hinders effectiveness. Even with policies in place, compliance and alignment issues with institutional priorities persist, compounded by insufficient leadership awareness and commitment. These issues highlight a critical gap in strategic anchoring, as IRs struggle to balance the expectations of external stakeholders, such as funders, with internal institutional priorities.

3.3 | Professionalizing

Forty-one articles in our sample of 50 articles (82%) mentioned challenges in terms of professionalizing governance and workflows, as well as staff capabilities. For

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this topic, we grouped coded segments for "Repository governance" and "Staff." Professionalization for IRs involves establishing clear governance structures, standardizing workflows, and ensuring that repository management aligns with best practices. This includes providing staff with the training, resources, and expertise required to effectively manage IR operations and adapt to evolving technological and institutional requirements.

One major challenge in the development of IRs is the lack of personnel in general (Anyaoku et al., 2019; Asadi et al., 2019; Doris & Ekereuche, 2023; Fasae et al., 2017; Joo et al., 2019; Kakai, 2021; Kipnis et al., 2019; Kyprianos & Lygnou, 2022; Martin-Yeboah et al., 2018; Oberhiri-Orumah & Baro, 2023; Shajitha & Majeed, 2021; Tapfuma & Hoskins, 2019; Ten Holter, 2020; Ullah & Mirza, 2020). For example, in a survey among Nigerian universities, 88% of respondents strongly agreed and agreed that the lack of ICT personnel was one of the challenges facing the development of IRs in Nigeria (Oberhiri-Orumah & Baro, 2023). In another larger survey conducted in the US in 2019, the lack of technical personnel was ranked second highest (Joo et al., 2019). In another study from 2022, participants identified the availability of staff with the skills and expertise needed as a major problem in Pakistan (Rafiq, 2022). There are many varying responsibilities in the management of IRs. It is difficult for the available personnel to find the time to handle all of them (Hwang et al., 2020; Kyprianos & Lygnou, 2022; Oladokun, 2015; Sterman, 2014; Tapfuma & Hoskins, 2019; Zhong & Jiang, 2016). This leads to the expansion of existing jobs, especially in cases where hiring new personnel is not possible for budget reasons (Abdelrahman, 2017; Tapfuma & Hoskins, 2019). This is either the case in settings with low budgets and staff in general, but also in countries like the UK, where libraries are thought to be better financed. Here, because of REF, the amount of deposited papers is quite large and a team of, for example, four people managing a repository in theory would be too small (Ten Holter, 2020).

Another challenge raised in the articles of our sample was insufficient training of existing personnel (Adam & Kiran, 2021; Adjei et al., 2019; Anyaoku et al., 2019; Asadi et al., 2019; Baro et al., 2014; Baro & Kari, 2016; Baro & Nwabueze-Echedom, 2023; Ifijeh et al., 2018; Kodua-Ntim & Fombad, 2020; Kyprianos & Lygnou, 2022; Mahanta, 2021; Okoroma, 2018; Rafiq, 2022; Scherer & Valen, 2019; Uba & Okonkwo, 2021; Ullah & Mirza, 2020; Zhong & Jiang, 2016). This is in part also the case because IR implementation and management includes responsibilities sometimes more akin to software and IT infrastructure work than "traditional" librarianship. For example, participants in one study described how it was hard to recruit people skilled in IT administration work for the IR due to the lower salary compared to private companies (Joo et al., 2019). At the same time, the skills needed for the development and maintenance of an IR are not always taught during librarian training (Rafiq, 2022). On the other hand, individuals who specialize in IT administration may lack the skills associated with traditional librarianship, and both parties do not always understand each other (Zhong & Jiang, 2016).

Furthermore, a dimension of professionalization challenges indicated in our sample was related to IR governance and the organization of repository services and workflows (Okoroma, 2018; Pinfield et al., 2014; Quinn, 2023; Rafiq, 2022; Sterman, 2014; Ten Holter, 2020; Utulu & Ngwenyama, 2021). Repository governance can be challenging due to complex decision-making processes in the overall RPO (i.e., a university), which can negatively influence the speed of implementation and spread of adoption of repositories (Pinfield et al., 2014). There may also be conflicting ideas present in terms of the implementation and scope of a repository system (Sterman, 2014). For example, conflicting implementation ideas between librarians about larger- and smaller-scale development projects for an IR were described in Utulu and Ngwenyama (2021). The same study also illustrated a conflicting view of how to deposit content into a repository between academics and IT staff in another setting. Additionally, views on the scope of what an IR should be used for can also conflict, especially when collection policies are very broad and poorly defined (Sterman, 2014). An example can be found in Liauw and Genoni (2017), whose study recounts that IRs in Indonesia were sometimes regarded by the university's leadership as an information management system for students' theses rather than a tool for scholarly communication.

Lastly, providing user-friendly services is another challenge in professionalization faced by repositories (Okoroma, 2018; Quinn, 2023; Rafiq, 2022). Our sample contained examples of challenges around inefficient services (Quinn, 2023; Ten Holter, 2020), mostly related to complicated user interfaces to deposit content in the repository. General service issues were also mentioned, describing the lack of opportunity for uploads but the need for researchers to submit files to librarians and IR managers (Okoroma, 2018). In addition to this, another study pointed to a challenge with general planning and procedures for repository services (Rafiq, 2022).

In summary, several challenges can be identified under the theme of professionalization. These particularly affect the management of IRs and are evident in aspects such as staffing and staff development. This includes the governance of IRs, with a special focus on

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the interaction between RPO as the hosting entity, libraries as the operating facilities, and in some cases, the interaction with IT departments within the RPO that manage the IRs. Challenges in the usability and efficiency of IRs are also apparent, especially concerning the use of IRs by researchers.

3.4 | Financing

Twenty-eight documents in our sample discussed challenges related to funding and financing (56%). For this theme, we analyzed all instances in which issues related to financial resources of the IR were discussed and subsequently coded with "Financing, funding." The distinction between financing aspects concerning the repository itself, the operating RPO, or broader economic conditions is not always clearly made in the reviewed literature. Thus, the distinction between financing and funding, the latter referring primarily to external resources from RFOs, is not clearly defined in the literature.

This challenge is particularly frequently mentioned in the set of articles focusing on repositories in Africa (Abdelrahman, 2017; Adam & Kiran, 2021; Adjei et al., 2019; Anyaoku et al., 2019; Baro et al., 2014; Baro & Kari, 2016; Baro & Nwabueze-Echedom, 2023; Dlamini & Snyman, 2017; Doris & Ekereuche, 2023; Ejikeme & Ifeanyi, 2019; Fasae et al., 2017; Ifijeh et al., 2018; Martin-Yeboah et al., 2018; Oberhiri-Orumah & Baro, 2023; Okoroma, 2018; Oladokun, 2015; Uba & Okonkwo, 2021; Ukwoma et al., 2019; Usman & Kaur, 2023; Utulu & Ngwenyama, 2021).

For example, of 25 IR managers in Nigeria, 100% of the respondents indicated that they strongly agreed or agreed that lack of funding was a significant challenge to the development of IRs in Nigeria (Oberhiri-Orumah & Baro, 2023). Similarly, in another survey 96% of 26 IR managers from various African RPOs, a lack of funding was selected as the most popular challenge for the development and use of IRs (Dlamini & Snyman, 2017). The authors of another study among library staff from Nigeria, Ghana and Uganda, which identified funding as the greatest obstacle to the development of IRs, stress that budget cuts across academic institutions in these countries resulted in this situation (Fasae et al., 2017). Okoroma (2018) emphasizes that "more financial interventions by the government and philanthropic organizations will result in a higher level of adoption and sustenance of IRs in Nigeria."

Studies from India and Pakistan also highlight financial challenges (Mahanta, 2021; Rafiq, 2022; Ullah & Mirza, 2020). In Mahanta's (Mahanta, 2021) survey among 126 librarians from Assam, India, financial problems are cited as the second most significant challenge. Similarly, Rafiq's (2022) study from Pakistan, which surveyed 34 university librarians, found that more than half of the respondents mentioned "financial issues" as affecting the operation of IRs.

The mention of economic aspects affecting the operation of IRs is by no means limited to the countries mentioned above. Studies from the US also address this topic. Sterman (2014), as a result of his survey of 26 IR managers in the US, describes the challenge of managing the IR without a dedicated budget or business plan. In the survey of 64 academic librarians responsible for managing an IR by Joo et al. (2019), it was found that the issue of a limited budget for systems and equipment presents a challenge for IR operations. Kipnis et al. (2019), in their survey of 50 members of the Association of Academic Health Sciences Libraries (AAHSL), identified funding issues and lack of human resources as affecting IRs in medical libraries in the US.

The fact that taking on new areas of work in IR operations also becomes a question of resources is made clear in the study by Ten Holter (2020) on the role of IRs in implementing the REF in the UK. The author highlights that the workload created by making REF articles accessible places significant demands on the resources of libraries and their IR departments. However, in a curious way, the topic of financial sustainability, specifically the importance of securing long-term funding for IRs to ensure the preservation of stored content, is explicitly mentioned in only five publications (Adjei et al., 2019; Baro & Kari, 2016; Rafiq, 2022; Sterman, 2014; Usman & Kaur, 2023).

In summary, funding IR presents a challenge that is clearly documented in the literature and is not limited to low-income and middle-income countries but remains difficult to define. Studies from the UK and the US indicate that funding issues are often linked with specific aspects that are not always detailed in these studies. A revealing look at the UK shows that with policy developments like the REF, the resources for IRs are also connected.

3.5 | Technology

Thirty-six articles (72%) illustrated challenges with Information and Communication Technology (ICT). Technological challenges are a persistent barrier to the development, implementation, and maintenance of IRs, particularly in resource-constrained environments. Issues such as unreliable internet connectivity, outdated and inadequate infrastructure, and limited technical expertise hinder the functionality and adoption of IRs. For this

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topic, we grouped segments coded with "Technology" and "Preservation."

The challenging domains that emerged under this topic especially concern internet connectivity and bandwidth issues (Asadi et al., 2019; Baro et al., 2014; Baro & Nwabueze-Echedom, 2023; Dlamini & Snyman, 2017; Doris & Ekereuche, 2023; Mahanta, 2021; Oberhiri-Orumah & Baro, 2023; Okoroma, 2018; Oladokun, 2015) and power supply issues (Adam & Kiran, 2021; Anyaoku et al., 2019; Asadi et al., 2019; Baro et al., 2014; Baro & Nwabueze-Echedom, 2023; Dlamini & Snyman, 2017; Fasae et al., 2017; Oberhiri-Orumah & Baro, 2023; Okoroma, 2018; Oladokun, 2015). It should be noted that these issues were not mentioned in articles with study samples in countries outside the African continent.

ICT setups for IRs are predominantly dependent on underlying infrastructures such as a reliable Internet connection and stable power supplies. However, several studies in our sample showed that both prerequisites are challenges in some settings. This is illustrated through the results of articles, which show that 65% (Dlamini & Snyman, 2017), 80% (Mahanta, 2021), 85% (Doris & Ekereuche, 2023), 92% (Oberhiri-Orumah & Baro, 2023), and up to 100% (Okoroma, 2018) among study participants agree that internet connectivity and bandwidth are lacking and hinder the development of IRs. The newest of these studies was published in 2023, so it seems like this is a challenge which persists to today. In the same vein, an unreliable and unpredictable power supply can be a fundamental challenge for IRs. This is mentioned in several articles, all of which originate from the African continent (Dlamini & Snyman, 2017; Fasae et al., 2017; Oberhiri-Orumah & Baro, 2023; Okoroma, 2018).

Lastly, technological challenges also present themselves in the form of a lack of ICT development that results in outdated or obsolete hardware and software, or complex to maintain in-house solutions adequate for the contextual circumstances (Adjei et al., 2019; Anyaoku et al., 2019; Asadi et al., 2019; Baro & Nwabueze-Echedom, 2023; Marsh, 2015; Okoroma, 2018; Scherer & Valen, 2019; Ullah & Mirza, 2020; Utulu & Ngwenyama, 2021; Zhong & Jiang, 2016). Examples include outdated software such as Windows XP instead of Windows 10 running on official computers (Okoroma, 2018) or self-made software for repositories that is not interoperable with more popular, but also complex solutions such as DSpace (Zhong & Jiang, 2016). Notably, challenges for preserving information in IRs were mostly mentioned in passing with other technological challenges, except for a study from India, in which 45.2% of surveyed participants selected preservation problems as one of the challenges to IR development (Mahanta, 2021).

In summary, challenges such as internet connectivity and bandwidth issues, as well as power supply issues, can be stated to be a central theme in African publications. However, ICT-related challenges are also clearly relevant beyond that, such as the currency of hardware and software environments for IR. These factors impact the persistence of publications made accessible through IR.

3.6 | Content and quality

Content and quality in the context of IRs refer to the scope, reliability, and management of materials stored within these systems. Content challenges often involve ensuring the inclusion of diverse and relevant research outputs while navigating restrictions such as copyright, embargoes, and inconsistent submission practices. Quality challenges pertain to maintaining high standards for metadata accuracy, accessibility, and usability to ensure discoverability. We grouped segments coded with "Data sharing," "Metdata," "Rights," and "Quality, reputation" for this theme. Thirty-seven documents (74%) in our sample included at least one of these codes.

One major challenge for IRs, which was discussed in 33 papers, revolves around copyright and secondary publication rights. Rights issues can be very complicated and hard to understand for authors and might discourage them from sharing their publications via an IR (Dawson & Yang, 2016; Dubinsky, 2014; Farida et al., 2015; Fasae et al., 2017; Ifijeh et al., 2018; Oberhiri-Orumah & Baro, 2023). This challenge is especially evident in cases where there is a stark absence of explicit copyright policies or at least information on handling copyright by the IR (Dawson & Yang, 2016; Ullah & Mirza, 2020; Zhong & Jiang, 2016) or when IR managers themselves are not sufficiently trained in copyright handling (Hwang et al., 2020; Kyprianos & Lygnou, 2022; Zhong & Jiang, 2016). Some repositories try to mitigate this challenge by integrating services like the SHERPA/RoMEO database (renamed to Open Policy Finder in October 2024) for copyright information (Farida et al., 2015; Kakai et al., 2018; Oberhiri-Orumah & Baro, 2023; Shajitha & Majeed, 2021).

The role of repositories in data sharing is addressed in seven papers, none of which originated from the African continent. It became evident that the publication type data was often not clearly defined, making it difficult to obtain reliable information on this topic. In our view, the term is used in some papers as an umbrella concept for all forms of publication types, while in others, the term research data is specifically employed, with a clear reference to FAIR and open research data principles.

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Quality control is another critical concern. Many repositories struggle with metadata accuracy and consistency, as well as ensuring that submissions adhere to defined quality standards. Insufficient resources for metadata creation and inconsistent practices across collections make it difficult to maintain interoperability and discoverability (Joo et al., 2019; Marsh, 2015). In some cases, librarians may not allow researchers to self-archive due to fear of a lack of quality of the metadata (Dubinsky, 2014). Issues such as missing or broken links, poorly classified content, and non-standard naming conventions further degrade the quality of repository content (Liauw & Genoni, 2017). To address these challenges, dedicated positions for librarians knowledgeable about metadata standards must be created (Tapfuma & Hoskins, 2019). However, due to connected financial and professionalization challenges, this cannot always be possible.

The perceived quality and reputation of IR content also pose challenges for IRs to further their adoption by researchers. Scholars, particularly in disciplines with established subject repositories, often perceive IRs as inferior due to a lack of peer review or lower visibility compared to other platforms (Dubinsky, 2014). Because many repositories do not provide clear usage metrics for their content to authors, it may seem to researchers that making the effort to publish their work in the IR would not benefit their academic careers, which are probably built on the scholarly reputation system and citation counts (Ten Holter, 2020). This skepticism extends to fears of plagiarism or misuse of shared materials, further discouraging participation (Uba & Okonkwo, 2021; Zhong & Jiang, 2016). In addition, in some regions, repositories prioritize internal or institutional goals, such as combating plagiarism, over broader open access objectives, which can limit their appeal and usability (Liauw & Genoni, 2017).

In summary, challenges related to copyright aspects, quality assurance, and the associated reputation of the IRs present issues that can be categorized under the "Content and Quality" theme. The literature indicates that uncertainties regarding legal or quality aspects affect the operation of the IRs and also influence how researchers perceive them, thereby impacting the usage of the repositories.

3.7 | Usage

IRs face numerous challenges that hinder their use and broader adoption, which can be categorized into aspects such as awareness and acceptance, content acquisition, visibility and discoverability, marketing, evaluation of usage, and usability issues. We grouped segments coded with "Usage, awareness," "Marketing, communication," and "Impact, statistics" under this topic. Thirty-seven documents (74%) contained these topics.

Lack of knowledge among researchers as authors and users of IRs, as well as among the leadership of RPOs, is identified as a challenge in many of the reviewed articles. Studies highlight that limited awareness among researchers and academics is a significant obstacle to their development and use (Abdelrahman, 2017; Anyaoku et al., 2019; Asadi et al., 2019; Dlamini & Snyman, 2017; Ifijeh et al., 2018; Marsh, 2015; Oberhiri-Orumah & Baro, 2023; Ten Holter, 2020; Zhong & Jiang, 2016). Additionally, the absence of awareness at the institutional management level further hinders progress (Ejikeme & Ifeanyi, 2019; Hwang et al., 2020; Rafiq, 2022). This lack of strategic integration within institutions also undermines the administrative support needed to advance IR initiatives.

The general apathy of scholars towards IRs, characterized by a failure to recognize their benefits, further exacerbates these challenges. Some studies emphasize that when faculty understand the advantages of IRs, such as increased visibility and accessibility of their work, their participation increases (Farida et al., 2015; Kakai, 2021; Ten Holter, 2020). Joo et al. (2019) highlight the potential of Search Engine Optimization (SEO) as a means to enhance the visibility of publications stored in IRs. However, usability issues can hinder discoverability, as Zhong and Jiang (2016) observe in their study of Chinese IRs, where features are often misleading and links or URLs do not connect to the correct content. Liauw and Genoni (2017) further emphasize the importance of English metadata for IRs in Indonesia, noting that it is critical to achieve global visibility and integration into the international scientific community.

However, challenges in content acquisition are still central to IRs and are discussed in many studies (Anyaoku et al., 2019; Baro & Nwabueze-Echedom, 2023; Doris & Ekereuche, 2023; Dubinsky, 2014; Farida et al., 2015; Joo et al., 2019; Kakai, 2021; Kodua-Ntim & Fombad, 2020; Oberhiri-Orumah & Baro, 2023; Oladokun, 2015; Pinfield et al., 2014; Quinn, 2023; Uba & Okonkwo, 2021). Oberhiri-Orumah and Baro (2023) note that collecting content for IRs is a significant challenge in Nigeria's IR development. Similarly, Doris and Ekereuche (2023) found that half of the repository managers surveyed reported low interest among contributors to submit their publications. They identified content acquisition as a major challenge for African IRs, attributed to factors such as unclear policies, concerns about the misuse of Open Access content, and author hesitations. Other reasons for these challenges are identified as faculty

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disinterest, lack of time, and unwillingness as key barriers to self-archiving (Dubinsky, 2014; Marsh, 2015; Oladokun, 2015) or copyright concerns, a preference for disciplinary repositories, perceived complexity in the submission process, and fears of plagiarism (Doris & Ekereuche, 2023; Joo et al., 2019). Joo et al. (2019) emphasize that many researchers are unaware of IRs and are reluctant to contribute their work. They attribute this to a lack of motivation and limited awareness of the potential benefits of sharing data and publications through IRs. To address these issues, they recommend workshops or materials that highlight the advantages of depositing works, alongside clear guidelines on rights, simplified copyright review processes, and instructions to make submissions easier.

This highlights the importance of targeted outreach, such as workshops, advocacy efforts, and training programs, to increase awareness and foster engagement. Kakai (2021) underscores the importance of active involvement from librarians, particularly through direct engagement with researchers, which has been shown to increase their willingness to contribute. Effective IRs should extend beyond library-centric models by involving stakeholders across campus. Additionally, the inclusion of research data as a publication type presents new challenges for IRs, as noted by Joo et al. (2019), Kodua-Ntim and Fombad (2020), and Pinfield et al. (2014).

Several studies identify inadequate marketing as a problem for IRs (Fasae et al., 2017; Hwang et al., 2020; Kakai et al., 2018; Martin-Yeboah et al., 2018; Oladokun, 2015). Hwang et al. (2020) note that marketing efforts are often directed at content providers rather than end-users, leaving IRs dependent on search engines for content discoverability. Meetings with individual faculty or small groups are identified as effective methods for acquiring content, yet active marketing remains underutilized. Farida et al. (2015) describe such efforts as the "socialization" (p. 176) of an IR, which can include formal and informal outreach through print, online, and in-person engagements. Martin-Yeboah et al. (2018) clearly states that a "'build it and they will come' attitude" (p. 17), which is pursued by some IRs, will not work. The importance of marketing also becomes relevant when considering the competition from academic social networks such as ResearchGate, which are mentioned in two of the studies reviewed (Kakai et al., 2018; Ten Holter, 2020).

Usage metrics are vital to demonstrate the value and impact of IRs. A recurring issue is the perception of IRs as "black holes," where deposited work seemingly disappears without feedback or recognition, which discourages participation (Ten Holter, 2020). Reliable and transparent metrics, such as access statistics and citation data, provide essential feedback to researchers, motivating them to contribute by showcasing the visibility and influence of their work (Chilimo, 2015; Martí-Lahera & Puerta-Díaz, 2019). Publicly accessible usage statistics and alternative metrics are seen as effective tools for fostering engagement and demonstrating the wider reach of repository content (Baughman et al., 2018; Kakai, 2021). By linking these metrics with tangible outcomes, such as increased citations and greater academic visibility, IRs can align with the research goals and promote stronger participation, ultimately improving their value within the academic ecosystem (Quinn, 2023; Ten Holter, 2020). However, the precision and usability of current metrics remain a challenge, with calls for improvement to ensure that these systems effectively support researchers and administrators alike (Baughman et al., 2018).

In summary, the relationship between researchers and IRs presents a challenge that can be described as complex. This is evident in aspects such as awareness of the IR and knowledge about its role in scholarly communication. The literature highlights a variety of suggestions to improve the interaction between IRs and researchers. Marketing and the use of metrics are two aspects that are of particular significance in this context.

4 | DISCUSSION

This systematic review identified a broad number of challenges for IRs from the literature. These challenges span strategic, technical, organizational, and operational dimensions.

A central observation is that the success of IRs is dependent on their ability to align with institutional and regional priorities while also responding to global trends in scholarly communication. Strategic anchoring, including the development of policies and mandates, emerges as a global challenge, although with significant variability in implementation and enforcement. This variability stresses the need for adaptive strategies that respect local contexts while fostering broader interoperability and collaboration. A misalignment between IR objectives and institutional policies can hinder the effective adoption and utilization of IRs. Moreover, the lack of clear mandates or mechanisms to monitor compliance further complicates the strategic anchoring of IRs within the academic ecosystem.

Professionalization presents another significant set of challenges, particularly in staffing, governance, and usability. The management of IRs often requires collaboration across diverse organizational units, including libraries and IT departments, yet these interactions can be fraught with tension or misalignment. Many

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institutions lack adequately trained personnel, leading to gaps in repository management. In addition, the divergence between traditional librarianship training and the technical expertise required for IRs creates additional stress. This highlights the need for structured governance frameworks and targeted staff development programs that address both technical and operational competencies. The challenges here are not just operational but point to deeper questions about the roles and responsibilities of IRs within RPOs. The results suggest the need for institutions to navigate the tension between traditional librarianship models and the technical demands of modern repositories.

The financial sustainability of IRs also emerges as a persistent concern. While funding challenges are often associated with low- and middle-income countries, they are by no means limited to these contexts. Even in high-income countries like the UK and the US, resource allocation for IRs is often tied to policy developments, such as the REF in the UK. However, the literature indicates that financial issues are not always explicitly detailed, leaving significant gaps in understanding how funding constraints impact specific aspects of IR operations.

Another key theme is issues related to content and quality that intersect with researchers' perceptions and usage of IRs. Concerns about copyright, metadata quality, and the reputational value of IRs influence both their operation and their adoption by researchers. The review shows that improving repository adoption is not just a matter of technical fixes but requires addressing perceptions of value, usability, and alignment with incentives in the academic system. This highlights the need for IRs to position themselves as essential stakeholders in scholarly communication. Increasing visibility and demonstrating value to researchers are still needed efforts to achieve.

The challenges facing IRs vary significantly between regions, reflecting differences in economic, technological, and institutional contexts. Although institutions in high-income countries often grapple with policy compliance and scaling, those in low-income and middle-income countries face foundational issues such as funding and infrastructure. Addressing these disparities requires a concerted global effort, including the sharing of best practices and resources through international collaborations.

The scope of our sample is significantly sloped towards articles from African countries. This might be due to the selected time period. It might be that there is an emphasis on topics such as unreliable technology that is over-represented in countries in which the development of IRs is slower than in countries with, for example, more reliable infrastructures. However, the topics discussed around these issues might have been relevant in the literature from countries such as the US or the UK before the time period we chose and are therefore not included in the sample.

Reflecting the literature on challenges for IRs as a whole, it became clear that while there is a large amount of literature on the usage or non-usage of IRs, the number of records that are concerned with challenges faced by the infrastructure is not very high. Put differently, the discourse around challenges connected to IRs is oftentimes more concerned with studies on usage or non-usage of IRs or challenges faced by scholars than with studies on internal challenges faced, for example, by IR managers.

4.1 | Limitations

There are several limitations to these results. First, the scope of the LISA database, which focuses primarily on library and information science, may have restricted the breadth of relevant literature included. In hindsight, we were not satisfied with the quality of the LISA results and the usability of the database. Alongside complicated retrieval of metadata and no possibility for batch downloads, unfortunately the metadata was also not curated very well. In our final sample of 50 articles, six articles contained faulty metadata fields, especially regarding the spelling of names and authority control issues, for example, differentiating first and last names. Additionally, the selection criteria for which journals are included in LISA are not quite comprehensive and should be made more transparent and probably updated. Expanding the search to an additional database (Scopus) after review was expected to capture a broader range of perspectives and disciplines by including articles not published in LIS-specific journals. However, all 16 additional articles included from Scopus were also published in LIS-specific journals.

Second, the selected time period for the review may have influenced the emphasis on certain topics, such as technology challenges, which are more prominent in regions where the development of IRs lags due to infrastructural limitations. As a result, discussions on similar issues in more developed contexts might have been published before the chosen time frame; they are underrepresented in this review. Future studies should address these limitations by incorporating more diverse databases and longer time frames to ensure a more comprehensive understanding of the challenges facing IRs around the world.

Third, while our study comprehensively analyzes a decade of research (2014–2024) on institutional

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repositories, we recognize that this field continues to evolve. The cutoff date for our literature search means that very recent developments may not be captured in our analysis. However, given the substantial timeframe covered and the patterns observed throughout this period, we believe our findings present a robust representation of the major trends and issues. Future research could valuably extend this work by examining how the landscape has continued to develop beyond our study period, particularly in response to emerging technologies and changing scholarly communication practices.

Finally, we did not conduct a formal critical appraisal or risk of bias assessment of included studies. We documented methodological characteristics, such as study design, methods, population, and location, to contextualize our findings, and through this process, we identified common methodological limitations across the literature, including small sample sizes and insufficient methodological detail in some studies, which may affect the generalizability and reproducibility of the findings. Future systematic reviews on IR challenges would benefit from incorporating quality assessment tools.

5 | CONCLUSION

This systematic review provides a comprehensive examination of the challenges facing IRs as documented in LIS research. Based on the analysis of 50 studies selected from an initial pool of 132,417 articles, we identified six core themes that encapsulate these challenges: Strategic anchoring, Professionalization, Financing, Technology, Content and quality, and Usage. These themes underscore the varied and interconnected nature of the challenges IRs encounter, reflecting both external influences, such as the Open Science movement and policies from RFOs, and internal dynamics rooted in institutional practices.

The findings reveal that IRs must navigate complex interactions with institutional and external policies, address persistent resource and staffing limitations, and adapt to the evolving technological landscape. Furthermore, the changing nature of scholarly outputs, with increasing emphasis on research data and software, requires IRs to expand their scope and functionality. Challenges related to copyright, quality assurance, and reputation, along with the need to improve researcher engagement through marketing and metrics, need to be addressed.

In summary, this review highlights the urgent need for holistic approaches to address these challenges, balancing the strategic, operational, and technical dimensions of IR management. By acknowledging the interplay of external and internal factors and striving for collaboration across stakeholders, IRs can position themselves as essential infrastructures in the global Open Science ecosystem.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the authors upon request.

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ENDNOTES

[1] See https://www.jisc.ac.uk/opendoar. Last checked Saturday August 09 11:25:48 2025.

[2] See https://about.proquest.com/en/products-services/lisa-set-c/. Last checked Saturday August 09 11:25:49 2025.

[3] See https://api.openalex.org/. Day of data retrieval for LISA records: 28 November 2024. Day of data retrieval for Scopus records: 17 June 2025.

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